COVID DATASET ANALYSIS

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DOMAIN: DATA SCIENCE

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covid dataset case study

[178]:
```
import pandas as pd
from matplotlib import pyplot as plt
import seaborn as sns
import datetime as dt
import numpy as np
```

[179]:
```
df=pd.read_csv('covid_19_india.csv',parse_dates=['Date'],dayfirst=True)
```

```
C:\Users\HP\AppData\Local\Temp\ipykernel_23956\689802896.py:1: UserWarning: Parsing dates in %Y/%m/%d format when dayfirst=Tru
first=False` or specify a format to silence this warning.
  df=pd.read_csv('covid_19_india.csv',parse_dates=['Date'],dayfirst=True)
```

[180]:
```
#looking at earluer dates
df.head()
```

[180]:

	Unnamed: 0	Date	State/UT	Latitude	Longitude	Confirmed	Active	Deaths	Mortality rate	Cured	Recovery rate
0	0	2020-01-30	Kerala	10.8505	76.2711	1	1	0	0.0	0	0.0
1	1	2020-01-31	Kerala	10.8505	76.2711	1	1	0	0.0	0	0.0
2	2	2020-02-01	Kerala	10.8505	76.2711	2	2	0	0.0	0	0.0
3	3	2020-02-02	Kerala	10.8505	76.2711	3	3	0	0.0	0	0.0
4	4	2020-02-03	Kerala	10.8505	76.2711	3	3	0	0.0	0	0.0

[181]:
```
#looking at the latest dates
df.tail()
```

[181]:

	Unnamed: 0	Date	State/UT	Latitude	Longitude	Confirmed	Active	Deaths	Mortality rate	Cured	Recovery rate
947	947	2020-04-15	Telengana	18.1124	79.0193	624	507	17	0.027244	100	0.160256
948	948	2020-04-15	Tripura	23.9408	91.9882	2	2	0	0.000000	0	0.000000
949	949	2020-04-15	Uttar Pradesh	26.8467	80.9462	660	605	5	0.007576	50	0.075758
950	950	2020-04-15	Uttarakhand	30.0668	79.0193	37	28	0	0.000000	9	0.243243
951	951	2020-04-15	West Bengal	22.9868	87.8550	213	169	7	0.032864	37	0.173709

[182]:
```
df=df[['Date','State/UT','Cured','Deaths','Confirmed']]
df.columns=['date','state','cured','deaths','confirmed']
```

HERE WE ARE RENAMING THE DATA AND CHOOSING ONLY REQUIRED DATA

[183]:
```
df.head()
df.tail()
```

[183]:

	date	state	cured	deaths	confirmed
947	2020-04-15	Telengana	100	17	624
948	2020-04-15	Tripura	0	0	2
949	2020-04-15	Uttar Pradesh	50	5	660
950	2020-04-15	Uttarakhand	9	0	37
951	2020-04-15	West Bengal	37	7	213

[184]:
```
today=df[df.date == '2020-04-15']
```

[185]:
```
today.head()
```

[185]:

	date	state	cured	deaths	confirmed
919	2020-04-15	Andaman and Nicobar Islands	10	0	11
920	2020-04-15	Andhra Pradesh	16	9	483
921	2020-04-15	Arunachal Pradesh	0	0	1
922	2020-04-15	Assam	0	1	32
923	2020-04-15	Bihar	29	1	66

[186]:
```
#sorting data wrt number of vonfirmed cases
max_confirmed_cases=today.sort_values(by='confirmed',ascending=False)
max_confirmed_cases
```

[186]:

	date	state	cured	deaths	confirmed
937	2020-04-15	Maharashtra	259	178	2687
926	2020-04-15	Delhi	30	30	1561
946	2020-04-15	Tamil Nadu	81	12	1204
945	2020-04-15	Rajasthan	147	3	969
936	2020-04-15	Madhya Pradesh	51	50	730
949	2020-04-15	Uttar Pradesh	50	5	660
928	2020-04-15	Gujarat	59	28	650
947	2020-04-15	Telengana	100	17	624
920	2020-04-15	Andhra Pradesh	16	9	483
934	2020-04-15	Kerala	211	3	387
931	2020-04-15	Jammu and Kashmir	30	4	278

HERE WE ARE GETTING THE MAXIMUM NUMBER OF CONFIRMED CASES

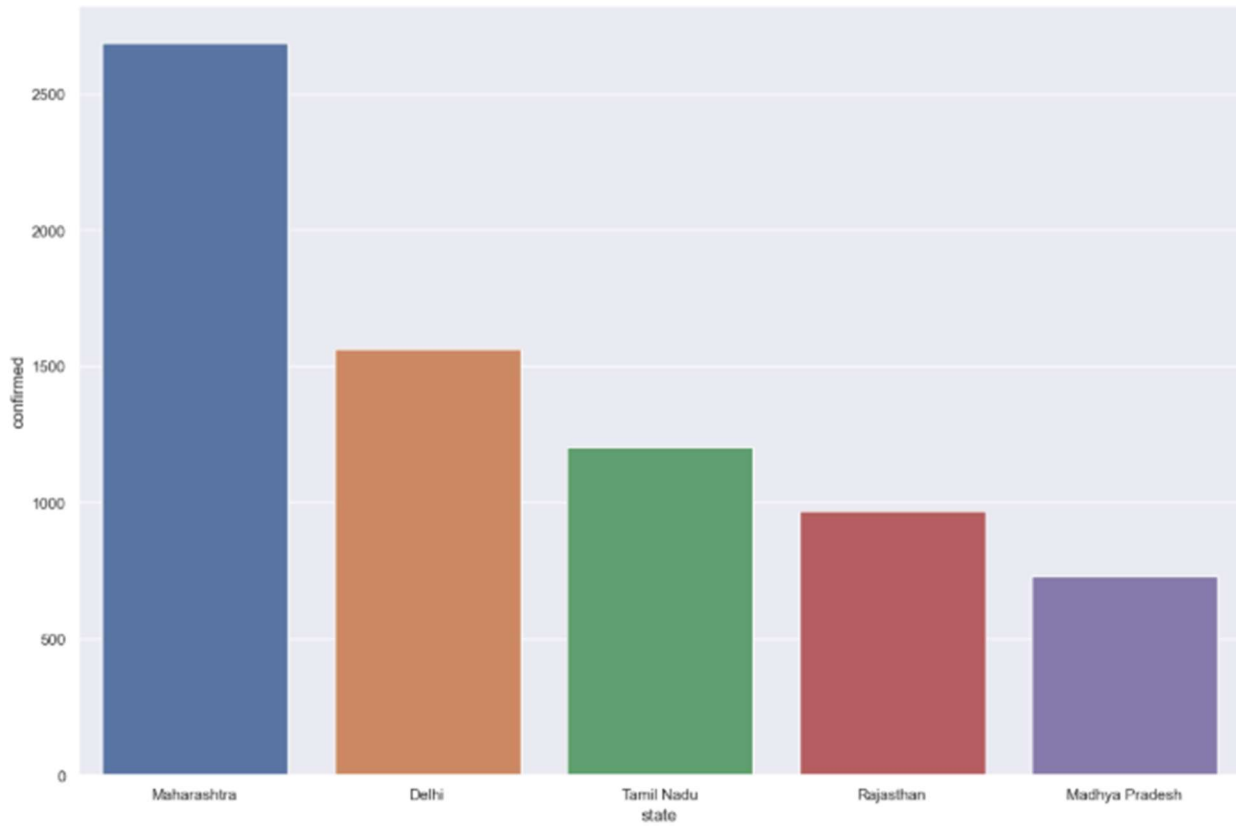
[187]:
```
top_states_confirmed=max_confirmed_cases[0:5]
```

[188]:
```
top_states_confirmed
```

[188]:

	date	state	cured	deaths	confirmed
937	2020-04-15	Maharashtra	259	178	2687
926	2020-04-15	Delhi	30	30	1561
946	2020-04-15	Tamil Nadu	81	12	1204
945	2020-04-15	Rajasthan	147	3	969
936	2020-04-15	Madhya Pradesh	51	50	730

[189]:
```
#marking bar plot for state with top confirmed cases
sns.set(rc={'figure.figsize':(15,10)})
sns.barplot(x="state",y="confirmed",data=top_states_confirmed,hue="state")
plt.show()
```



HERE WE ARE GETTING THE STATES WITH MAXIMUM CONFIRMED CASES AND PLOTTING A BAR PLOT OF TOP 5 STATES

```
[190]: #sorting data wrt number of death cases
max_death_cases=today.sort_values(by="deaths",ascending=False)
max_death_cases
```

[190]:

	date	state	cured	deaths	confirmed
937	2020-04-15	Maharashtra	259	178	2687
936	2020-04-15	Madhya Pradesh	51	50	730
926	2020-04-15	Delhi	30	30	1561
928	2020-04-15	Gujarat	59	28	650
947	2020-04-15	Telengana	100	17	624
946	2020-04-15	Tamil Nadu	81	12	1204
944	2020-04-15	Punjab	14	12	176
933	2020-04-15	Karnataka	71	10	260
920	2020-04-15	Andhra Pradesh	16	9	483
951	2020-04-15	West Bengal	37	7	213
949	2020-04-15	Uttar Pradesh	50	5	660
931	2020-04-15	Jammu and Kashmir	30	4	278
945	2020-04-15	Rajasthan	147	3	969
934	2020-04-15	Kerala	211	3	387
929	2020-04-15	Haryana	34	3	199
932	2020-04-15	Jharkhand	0	2	27
942	2020-04-15	Odisha	18	1	60
923	2020-04-15	Bihar	29	1	66
930	2020-04-15	Himachal Pradesh	13	1	33
922	2020-04-15	Assam	0	1	32
919	2020-04-15	Andaman and Nicobar Islands	10	0	11
921	2020-04-15	Arunachal Pradesh	0	0	1
924	2020-04-15	Chandigarh	7	0	21
925	2020-04-15	Chhattisgarh	13	0	33
927	2020-04-15	Goa	5	0	7
935	2020-04-15	Ladakh	10	0	17
943	2020-04-15	Puducherry	1	0	7
940	2020-04-15	Mizoram	0	0	1
939	2020-04-15	Meghalaya	0	0	1

930	2020-04-15	Himachal Pradesh	13	1	33
922	2020-04-15	Assam	0	1	32
919	2020-04-15	Andaman and Nicobar Islands	10	0	11
921	2020-04-15	Arunachal Pradesh	0	0	1
924	2020-04-15	Chandigarh	7	0	21
925	2020-04-15	Chhattisgarh	13	0	33
927	2020-04-15	Goa	5	0	7
935	2020-04-15	Ladakh	10	0	17
943	2020-04-15	Puducherry	1	0	7
940	2020-04-15	Mizoram	0	0	1
939	2020-04-15	Meghalaya	0	0	1

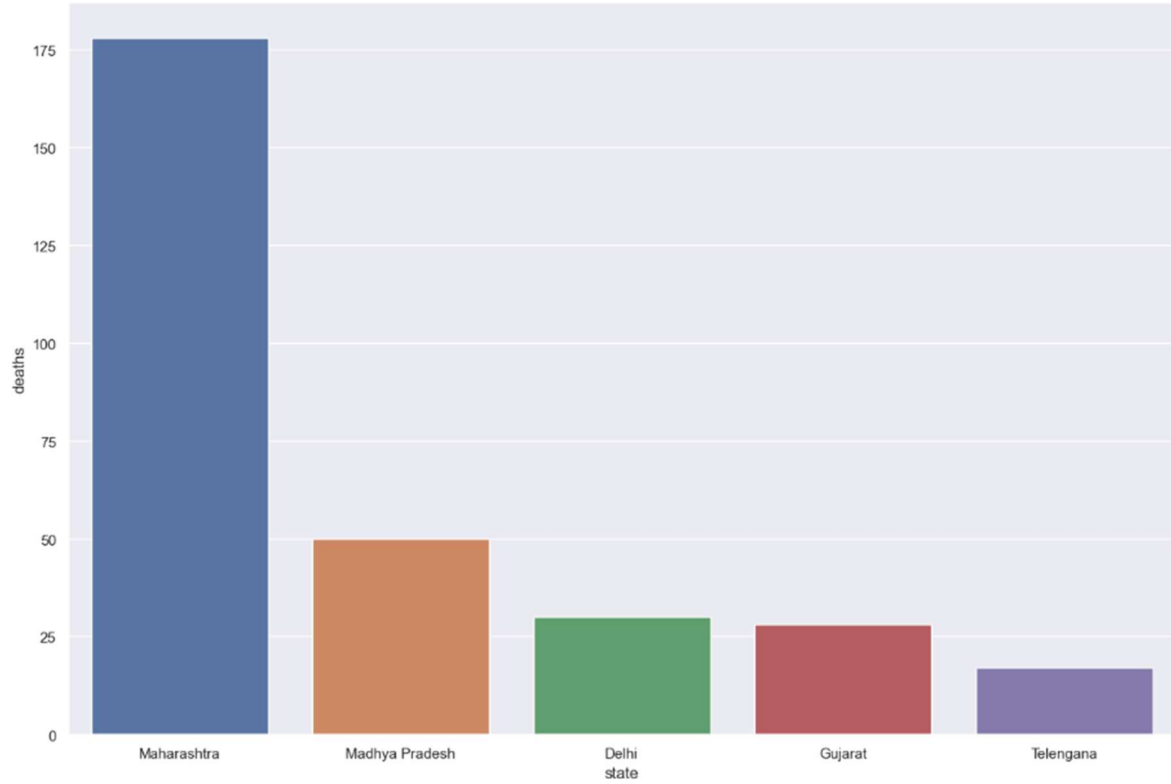
938	2020-04-15	Manipur	1	0	2
941	2020-04-15	Nagaland	0	0	0
948	2020-04-15	Tripura	0	0	2
950	2020-04-15	Uttarakhand	9	0	37

HERE WE FIND THE DEATH CASES

```
[214]: top_states_death=max_death_cases[0:5]
```

bar plot for states with top death cases

```
[215]: sns.set(rc={'figure.figsize':(15,10)})
sns.barplot(x="state",y="deaths",data=top_states_death,hue="state")
plt.show()
```



sorting data wrt number of cured cases

Here states with maximum death cases is taken and here are the top 5 states

sorting data wrt number of cured cases

```
max_cured_cases=today.sort_values(by="cured",ascending=False)
max_cured_cases
```

	date	state	cured	deaths	confirmed
937	2020-04-15	Maharashtra	259	178	2687
934	2020-04-15	Kerala	211	3	387
945	2020-04-15	Rajasthan	147	3	969
947	2020-04-15	Telengana	100	17	624
946	2020-04-15	Tamil Nadu	81	12	1204
933	2020-04-15	Karnataka	71	10	260
928	2020-04-15	Gujarat	59	28	650
936	2020-04-15	Madhya Pradesh	51	50	730
949	2020-04-15	Uttar Pradesh	50	5	660
951	2020-04-15	West Bengal	37	7	213
929	2020-04-15	Haryana	34	3	199
931	2020-04-15	Jammu and Kashmir	30	4	278
926	2020-04-15	Delhi	30	30	1561
923	2020-04-15	Bihar	29	1	66
942	2020-04-15	Odisha	18	1	60
920	2020-04-15	Andhra Pradesh	16	9	483
944	2020-04-15	Punjab	14	12	176
930	2020-04-15	Himachal Pradesh	13	1	33
925	2020-04-15	Chhattisgarh	13	0	33
919	2020-04-15	Andaman and Nicobar Islands	10	0	11
935	2020-04-15	Ladakh	10	0	17
950	2020-04-15	Uttarakhand	9	0	37
924	2020-04-15	Chandigarh	7	0	21
927	2020-04-15	Goa	5	0	7
938	2020-04-15	Manipur	1	0	2
943	2020-04-15	Puducherry	1	0	7
932	2020-04-15	Jharkhand	0	2	27
922	2020-04-15	Assam	0	1	32
921	2020-04-15	Arunachal Pradesh	0	0	1
940	2020-04-15	Mizoram	0	0	1
941	2020-04-15	Nagaland	0	0	0
939	2020-04-15	Meghalaya	0	0	1
948	2020-04-15	Tripura	0	0	2

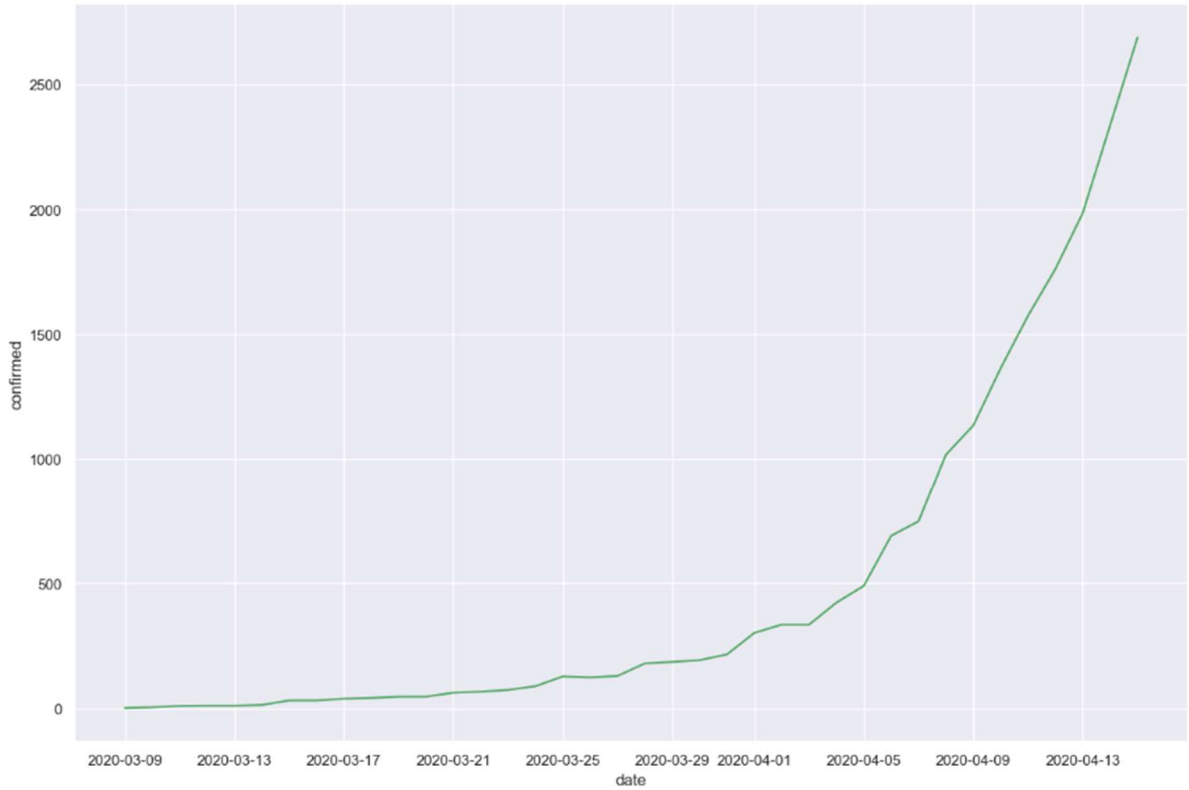
Now we see maharastas data

```
maha=df[df.state == 'Maharashtra']
maha
```

	date	state	cured	deaths	confirmed
77	2020-03-09	Maharashtra	0	0	2
89	2020-03-10	Maharashtra	0	0	5
101	2020-03-11	Maharashtra	0	0	10
114	2020-03-12	Maharashtra	0	0	11
127	2020-03-13	Maharashtra	0	0	11
140	2020-03-14	Maharashtra	0	0	14
153	2020-03-15	Maharashtra	0	0	32
167	2020-03-16	Maharashtra	0	0	32
182	2020-03-17	Maharashtra	0	1	39
197	2020-03-18	Maharashtra	0	1	42
215	2020-03-19	Maharashtra	0	1	47
235	2020-03-20	Maharashtra	0	1	47
260	2020-03-21	Maharashtra	0	1	63
283	2020-03-22	Maharashtra	0	2	67
306	2020-03-23	Maharashtra	0	2	74
329	2020-03-24	Maharashtra	0	2	89
353	2020-03-25	Maharashtra	1	3	128
380	2020-03-26	Maharashtra	1	3	124
407	2020-03-27	Maharashtra	15	4	130
434	2020-03-28	Maharashtra	25	5	180
461	2020-03-29	Maharashtra	25	6	186
488	2020-03-30	Maharashtra	25	8	193
515	2020-03-31	Maharashtra	39	9	216
544	2020-04-01	Maharashtra	39	9	302
572	2020-04-02	Maharashtra	42	13	335
601	2020-04-03	Maharashtra	42	16	335
631	2020-04-04	Maharashtra	42	19	423
660	2020-04-05	Maharashtra	42	24	490
690	2020-04-06	Maharashtra	42	45	690
720	2020-04-07	Maharashtra	56	45	748
751	2020-04-08	Maharashtra	79	64	1018
782	2020-04-09	Maharashtra	117	72	1135
813	2020-04-10	Maharashtra	125	97	1364
844	2020-04-11	Maharashtra	188	110	1574
875	2020-04-12	Maharashtra	208	127	1761
906	2020-04-13	Maharashtra	217	149	1985
937	2020-04-15	Maharashtra	259	178	2687

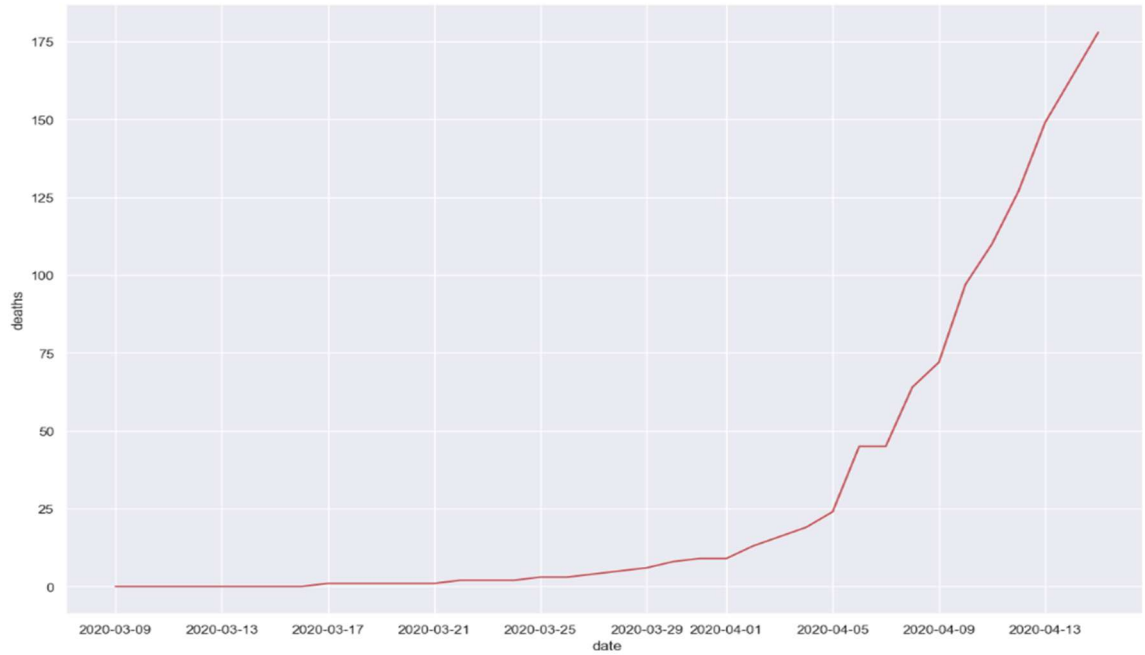
visualise confirmed cases in maharastra

195]:
```
sns.set(rc={'figure.figsize':(15,10)})
sns.lineplot(x="date",y="confirmed",data=maha,color="g")
plt.show()
```



visualising the death case in maharastra

[196]:
```
sns.set(rc={'figure.figsize':(15,10)})
sns.lineplot(x="date",y="deaths",data=maha,color="r")
plt.show()
```



Kerala

[197]:
```
kerala=df[df.state == 'Kerala']
kerala
```

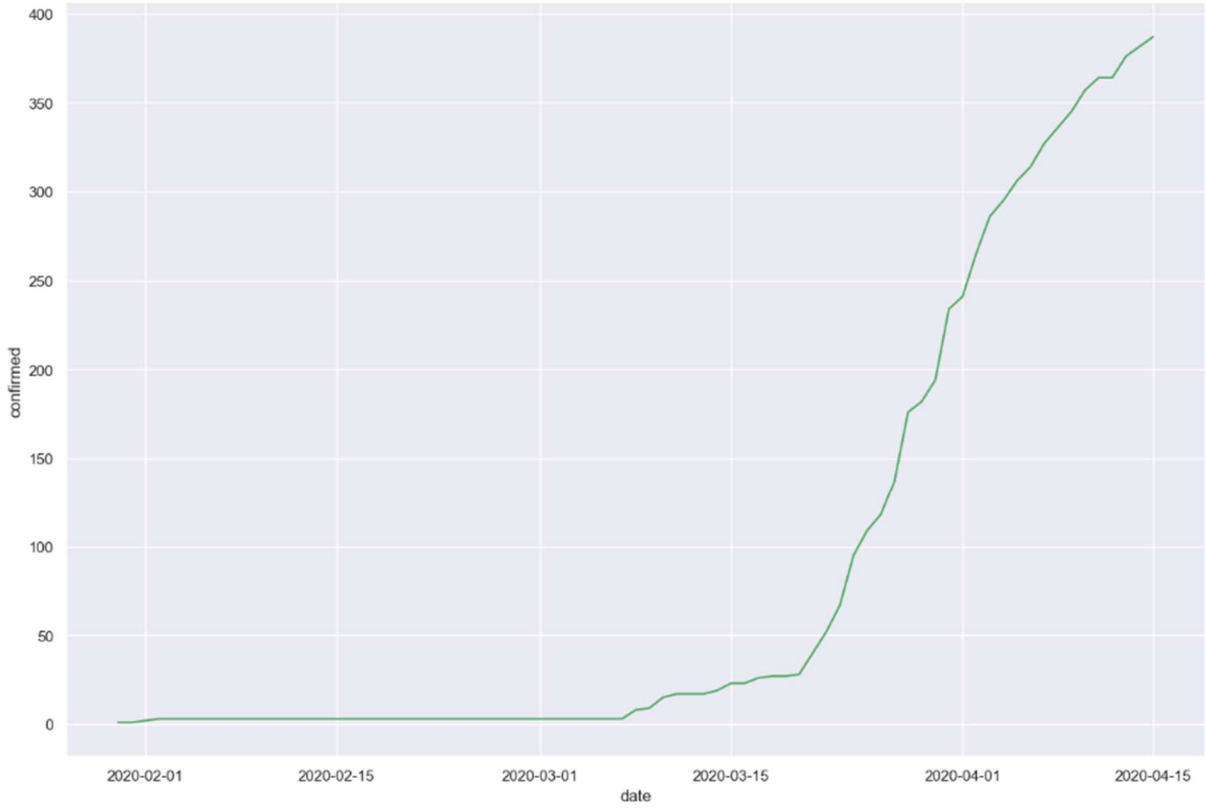
[197]:

	date	state	cured	deaths	confirmed
0	2020-01-30	Kerala	0	0	1
1	2020-01-31	Kerala	0	0	1
2	2020-02-01	Kerala	0	0	2
3	2020-02-02	Kerala	0	0	3
4	2020-02-03	Kerala	0	0	3
...	...	...	...	...	...
810	2020-04-10	Kerala	96	2	357
841	2020-04-11	Kerala	123	2	364
872	2020-04-12	Kerala	123	2	364
903	2020-04-13	Kerala	179	2	376
934	2020-04-15	Kerala	211	3	387

76 rows × 5 columns

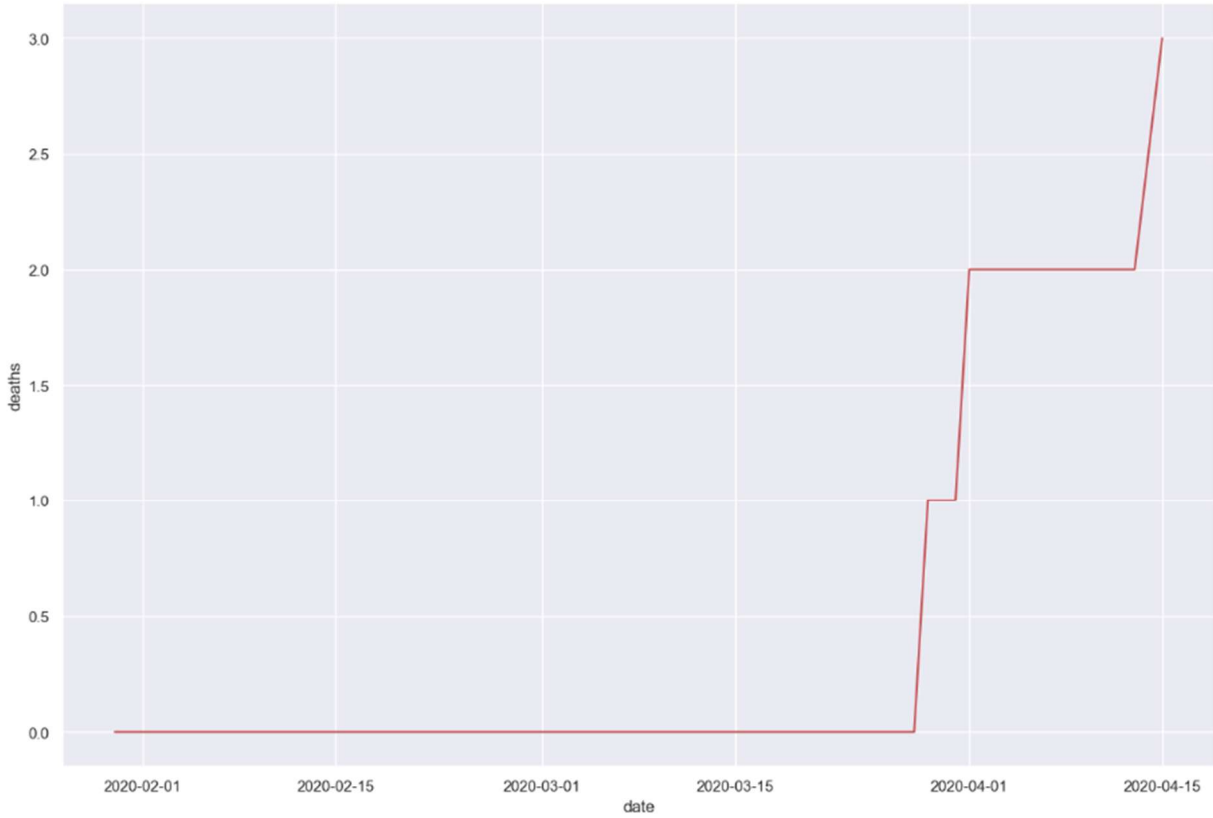
visualise confirmed case in kerala

[8]:
```
sns.set(rc={'figure.figsize':(15,10)})
sns.lineplot(x="date",y="confirmed",data=kerala,color="g")
plt.show()
```



visualise death case in kerala

```
sns.set(rc={'figure.figsize':(15,10)})
sns.lineplot(x="date",y="deaths",data=kerala,color="r")
plt.show()
```



Jammu and kashmir

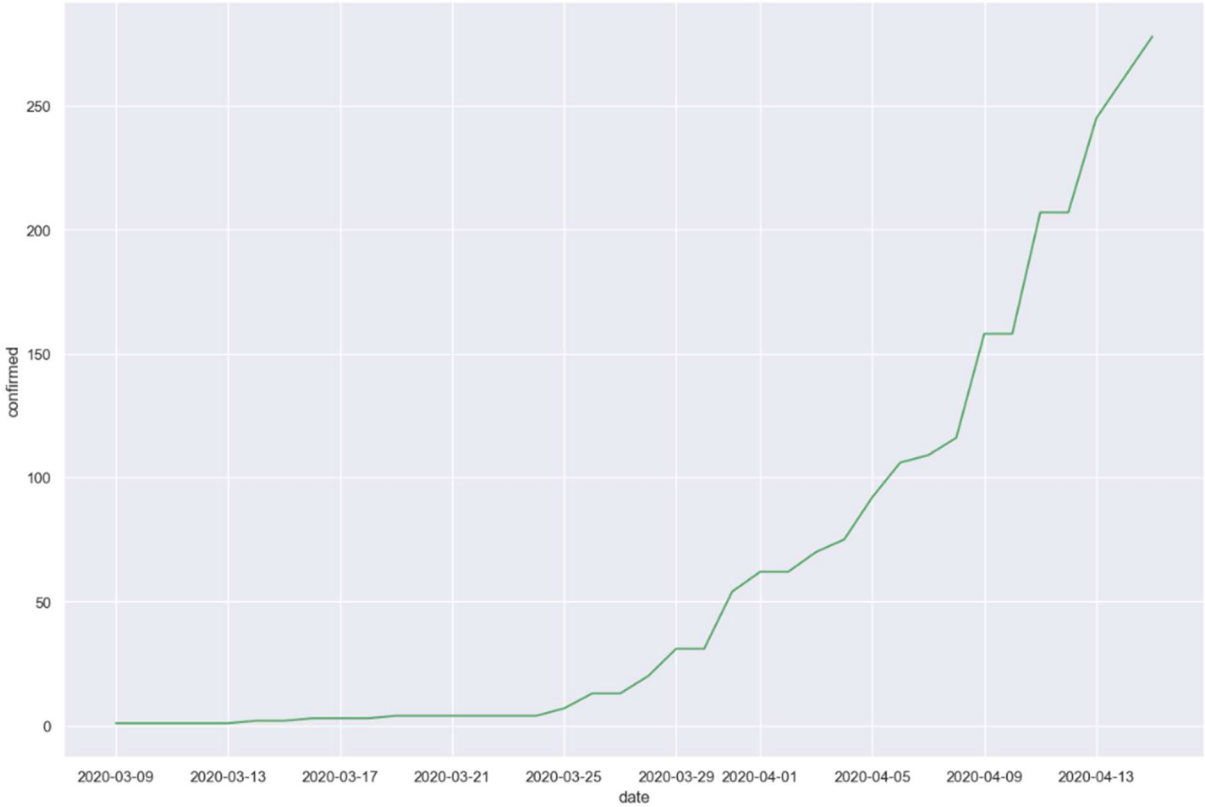
```
jk=df[df.state == 'Jammu and Kashmir']
jk
```

This is of jammu and Kashmir

	date	state	cured	deaths	confirmed
82	2020-03-09	Jammu and Kashmir	0	0	1
94	2020-03-10	Jammu and Kashmir	0	0	1
106	2020-03-11	Jammu and Kashmir	0	0	1
119	2020-03-12	Jammu and Kashmir	0	0	1
132	2020-03-13	Jammu and Kashmir	0	0	1
145	2020-03-14	Jammu and Kashmir	0	0	2
158	2020-03-15	Jammu and Kashmir	0	0	2
173	2020-03-16	Jammu and Kashmir	0	0	3
188	2020-03-17	Jammu and Kashmir	0	0	3
204	2020-03-18	Jammu and Kashmir	0	0	3
223	2020-03-19	Jammu and Kashmir	0	0	4
243	2020-03-20	Jammu and Kashmir	0	0	4
255	2020-03-21	Jammu and Kashmir	0	0	4
278	2020-03-22	Jammu and Kashmir	0	0	4
301	2020-03-23	Jammu and Kashmir	0	0	4
324	2020-03-24	Jammu and Kashmir	0	0	4
348	2020-03-25	Jammu and Kashmir	1	0	7
375	2020-03-26	Jammu and Kashmir	1	1	13
402	2020-03-27	Jammu and Kashmir	1	1	13
429	2020-03-28	Jammu and Kashmir	1	1	20
456	2020-03-29	Jammu and Kashmir	1	1	31
483	2020-03-30	Jammu and Kashmir	1	2	31
510	2020-03-31	Jammu and Kashmir	2	2	54
538	2020-04-01	Jammu and Kashmir	2	2	62
566	2020-04-02	Jammu and Kashmir	2	2	62
595	2020-04-03	Jammu and Kashmir	3	2	70
625	2020-04-04	Jammu and Kashmir	3	2	75
654	2020-04-05	Jammu and Kashmir	4	2	92
684	2020-04-06	Jammu and Kashmir	4	2	106
714	2020-04-07	Jammu and Kashmir	4	2	109
745	2020-04-08	Jammu and Kashmir	4	2	116
776	2020-04-09	Jammu and Kashmir	4	4	158
807	2020-04-10	Jammu and Kashmir	4	4	158
838	2020-04-11	Jammu and Kashmir	6	4	207
869	2020-04-12	Jammu and Kashmir	6	4	207
900	2020-04-13	Jammu and Kashmir	6	4	245
931	2020-04-15	Jammu and Kashmir	30	4	278

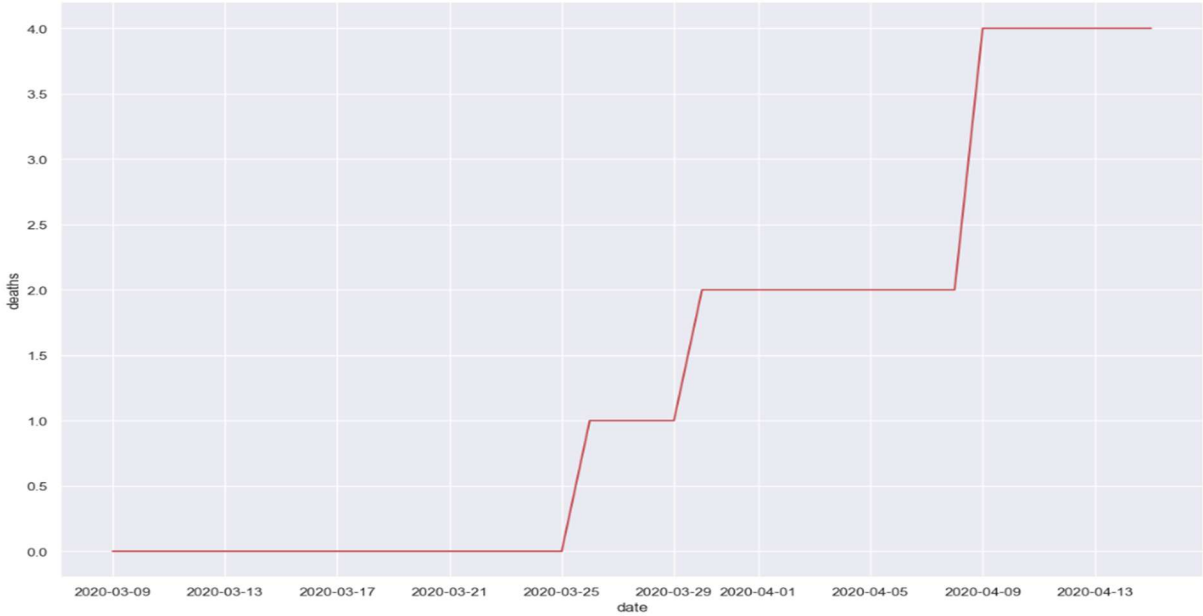
visualise confirmed case in JK

```
sns.set(rc={'figure.figsize':(15,10)})
sns.lineplot(x="date",y="confirmed",data=jk,color="g")
plt.show()
```



visualise death case in JK

```
sns.set(rc={'figure.figsize':(15,10)})
sns.lineplot(x="date",y="deaths",data=jk,color="r")
plt.show()
```



convert date time to ordinal

```
[3]: from sklearn.model_selection import train_test_split
```

```
[4]: maha['date']=maha['date'].map(dt.datetime.toordinal)
     maha.head()
```

```
C:\Users\HP\AppData\Local\Temp\ipykernel_23956\4140285815.py:1: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy
  maha['date']=maha['date'].map(dt.datetime.toordinal)
```

[4]:

	date	state	cured	deaths	confirmed
77	737493	Maharashtra	0	0	2
89	737494	Maharashtra	0	0	5
101	737495	Maharashtra	0	0	10
114	737496	Maharashtra	0	0	11
127	737497	Maharashtra	0	0	11

```
[5]: x=maha['cured']
     y=maha['confirmed']
```

```
[6]: X_train,x_test,y_train,y_test=train_test_split(x,y,test_size=0.3)
```

```
[7]: y_train
```

```
[7]: 937    2687
     660     490
     380     124
     720     748
     813    1364
     167      32
     461     186
     488     193
     690     690
     631     423
     77        2
     572     335
     751    1018
     544     302
     182      39
     101      10
     906    1985
     114      11
     215      47
     140      14
     306      74
     434     180
     283      67
     329      89
     235      47
     Name: confirmed, dtype: int64
```

Here we take independent variable to check how cure and confirmed are related

[208]:
```
from sklearn.linear_model import LinearRegression
dtr = LinearRegression()
dtr.fit(np.array(X_train).reshape(-1,1),np.array(y_train).reshape(-1,1))
```

[208]:
```
▾ LinearRegression
LinearRegression()
```

[209]:
```
maha.tail()
```

[209]:

	date	state	cured	deaths	confirmed
813	737525	Maharashtra	125	97	1364
844	737526	Maharashtra	188	110	1574
875	737527	Maharashtra	208	127	1761
906	737528	Maharashtra	217	149	1985
937	737530	Maharashtra	259	178	2687

[210]:
```
dtr.predict(np.array([[737625]]))
```

[210]:
```
array([[7315530.45036162]])
```

[211]:
```
dtr.predict(np.array([[2020-5-14]]))
```

[211]:
```
array([[19887.20595323]])
```